Aritra Dutta

Full Stack Application Engineer

+916295699190 aritradutta049@gmail.com Leetcode LinkedIn GitHub Website Bhubaneswar, India

PROFESSIONAL SUMMARY

Results-driven Full Stack Application Engineer with **2+** years of experience building scalable Java microservices, end-to-end web applications, and AI-powered features. Proficient in Java, Spring Boot, Python, Generative AI (RAG/LangChain), React, and AWS — with a strong focus on backend modernization and intelligent system integration, delivering measurable performance improvements in enterprise, client-facing environments.

WORK EXPERIENCES

System Engineer | Full-time — Sep 2024 - Present
Tata Consultancy Services — *On-site · Bhubaneswar, Odisha*

- Led the remediation phase of a Java microservices modernization initiative, migrating backend services from Java 8 to Java 21 and Spring Boot 1.x.x to 4.x.x, improving **maintainability, security, and long-term support**.
- Migrated critical applications from legacy JBoss infrastructure to Azure Tomcat PaaS, configuring health checks, deployment slots, and monitoring to ensure **high availability in production**.
- Utilized AWS Transform (ATX) tooling to accelerate dependency upgrades, code refactoring, and compatibility validation across Java microservices running on AWS-based infrastructure.
- Built and maintained end-to-end Jenkins CI/CD pipelines with Docker-based deployments, automated build and release workflows, integrated JaCoCo/JUnit test coverage reporting, and enhanced code quality through **SonarQube analysis**.
- Optimized complex SQL queries and REST APIs, delivering **up to 20× faster response times** for critical business workflows in production environments.
- Managed Git and GitHub workflows, including complex merge conflict resolution, pull request reviews, branch management, release merges, and enforcement of code review best practices across distributed development teams.
- Collaborated with UI/UX teams to implement and refine user-facing features, ensuring responsive design, usability improvements, and alignment with business requirements.
- Created, configured, and monitored Jenkins build and deployment jobs, troubleshooting pipeline failures and supporting continuous delivery practices across multiple environments.
- Conducted API validation and integration testing using Postman and Swagger/OpenAPI, partnering with QA teams to verify backend functionality, troubleshoot issues, and ensure release readiness.
- Leveraged GitHub Copilot and Claude Code to accelerate development, refactoring, documentation, and peer-review activities, improving **engineering productivity** across modernization initiatives.

Technologies / Skills Used : *Java, Spring Boot, Azure, AWS, Jenkins, Docker, SQL, React, AWS Transform, CICD, Tomcat, Node.js, Claude Code, Github Copilot*

PROJECTS

AlgoGuru | Github Live

- Designed, built, and deployed a full-stack competitive programming learning platform end-to-end, including production deployment and custom domain configuration.
- Integrated a **Generative AI-powered Guru Bot** to deliver context-aware algorithm explanations and intelligent, document-grounded problem-solving assistance.
- Built a Java-based code playground with role-based authentication, user progress tracking, and interactive problem sets using **React, TypeScript, and Tailwind CSS**.
- Owned the complete product lifecycle — architecture, AI integration, implementation, deployment, and ongoing maintenance.

Technologies / Tools Used : *React, TypeScript, Tailwind CSS, Java, Python, LangChain, RAG, Supabase, Node.js*

SKILLS

Databases :	PostgreSQL, Oracle, MySQL, Apache Kafka
Frameworks & Libraries :	Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, REST APIs, Microservices, LangChain, React, Angular, Tailwind CSS
Languages :	English, Hindi, Bengali
Programming Languages :	Java, Python, TypeScript, SQL
Soft Skills :	Agile, Scrum, Jira, Jile, Code Review
Tools & Platforms :	AWS, AWS Transform, Azure PaaS, Docker, Jenkins, Git, Maven, SonarQube, Tomcat, JBoss, GitHub Copilot, CI/CD

EDUCATIONS

Maulana Abul Kalam Azad University of Technology | Kolkata, India — 2020 - 2024
B.Tech — *CGPA : 8.79/10*

AWARDS & ACHIEVEMENTS

- Solved **700+** problems across competitive programming platforms — LeetCode (**1672**), Codeforces (**1046**), CodeChef (**1708**, **3-Star**), and CSES Problem Set.